

Principal Components Analysis (PCA)

Machine Learning Lecture Slides

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Prerequisites

Span of a set of vectors:

- For a vector space e.g. $\mathbf{R}^2$ or in general $\mathbf{R}^n$:
- the **span** of a set vectors e.g. $\mathbf{x} = \langle 1, 0 \rangle$, $\mathbf{y} = \langle 0, 1 \rangle$ is the set of all vectors $\mathbf{v}$ in $\mathbf{R}^2$ can be expressed as a linear combination:

$$\mathbf{v} = c_1 \mathbf{x} + c_2 \mathbf{y}$$

- In this example, the vectors $\mathbf{x}$, $\mathbf{y}$ span the whole $\mathbf{R}^2$
- Furthermore, $\mathbf{x}$, $\mathbf{y}$ is the minimal set to span the whole of $\mathbf{R}^2$
- In this case, $\mathbf{x}$, $\mathbf{y}$ form **bases** of $\mathbf{R}^2$

Prerequisites

Orthonormal vectors as bases:

- For a vector space, e.g. $\mathbf{R}^2$, we say that a vector $\mathbf{v}_1$ is *orthogonal* to another vector $\mathbf{v}_2$ if the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$
- Furthermore, we say that $\mathbf{v}_1$ and $\mathbf{v}_2$ are *orthonormal* if the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$ and $\|\mathbf{v}_1\| = \|\mathbf{v}_2\| = 1$
- For example, $\mathbf{x} = \langle 1, 0 \rangle$, $\mathbf{y} = \langle 0, 1 \rangle$
- Another example, $\mathbf{x} = \langle 1/\sqrt{2}, 1/\sqrt{2} \rangle$, $\mathbf{y} = \langle -1/\sqrt{2}, 1/\sqrt{2} \rangle$
- For $\mathbf{R}^2$ any two orthonormal vectors will span $\mathbf{R}^2$ and hence will form the bases
- Similarly, for $\mathbf{R}^n$ any n orthonormal vectors will span $\mathbf{R}^n$ and hence will form the bases

Prerequisites

Definition (Orthonormal matrix):

W is an orthonormal matrix if:

- $W = [w_1, \dots, w_n]$ with each w_i a column vector such that:
 - $w_i^T w_i = 1$ for any i i.e. w_i has unit length (note $w_i^T w_i$ is *dot product*)
 - $w_i^T w_j = 0$ for any $i \neq j$ i.e. w_i orthogonal to any other w_j
- Then it follows that $W W^T = I = W^T W$
- Hence, $W^{-1} = W^T$
- Any orthonormal basis transformation in $\mathbf{R}^n$ can be viewed as a **unitary transformation** (that includes, *rotation, reflection* and *combinations*)
- The purpose of dimensionality reduction such as PCA is to learn this transformation matrix from data *satisfying some additional properties*

Principal Component analysis

Aim: To find a new basis transformation such that:

$$\mathbf{x}_{new} = \mathbf{x}_{old} \mathbf{W}$$

where:

$\mathbf{x}_{new}$ is the old ($\mathbf{D} \times \mathbf{1}$) data item/row and $\mathbf{x}_{old}$ is the new transformed data item and $\mathbf{W}$ is a $\mathbf{D} \times \mathbf{D}$ transformation matrix.

Later on $\mathbf{W}$ will become a $\mathbf{D} \times \mathbf{L}$ (with $\mathbf{L} < \mathbf{D}$) transformation matrix.

Additionally $\mathbf{W}$ needs to satisfy:

1) Each column of $\mathbf{W}$ corresponds to a basis vector such that:

$\mathbf{w}_i \cdot \mathbf{w}_j$ (dot product) = 0 for $i \neq j$

$|| \mathbf{w}_i || = 1$ for any $\mathbf{w}_i$

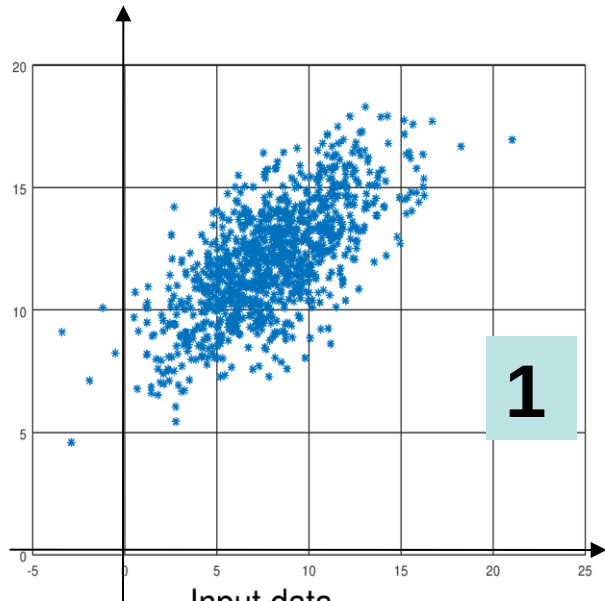
Or in other words:

$\mathbf{W}$ is an orthonormal basis i.e. $\mathbf{W}^T \mathbf{W} = \mathbf{I}$

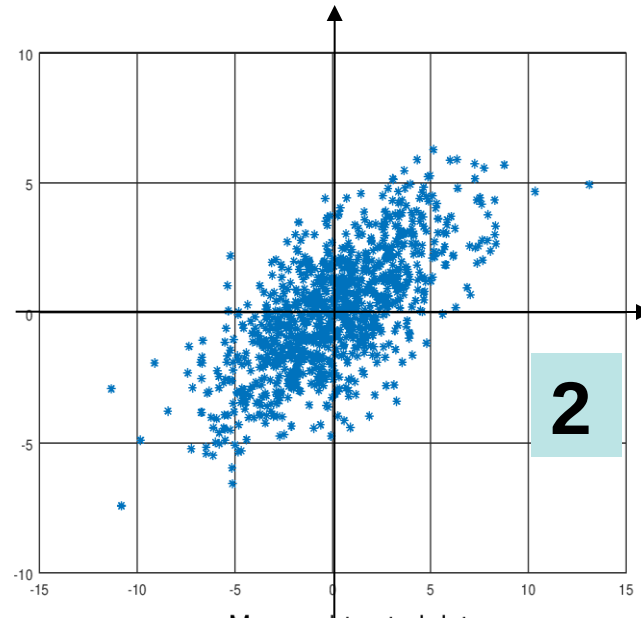
1) $\mathbf{w}_1$ gives maximum variance:

$$\mathbf{w}_1^* = \underset{||\mathbf{w}_1||=1}{arg\ max} \left[\sum_{i=1}^N x_{i_{new}}^2 \right]$$

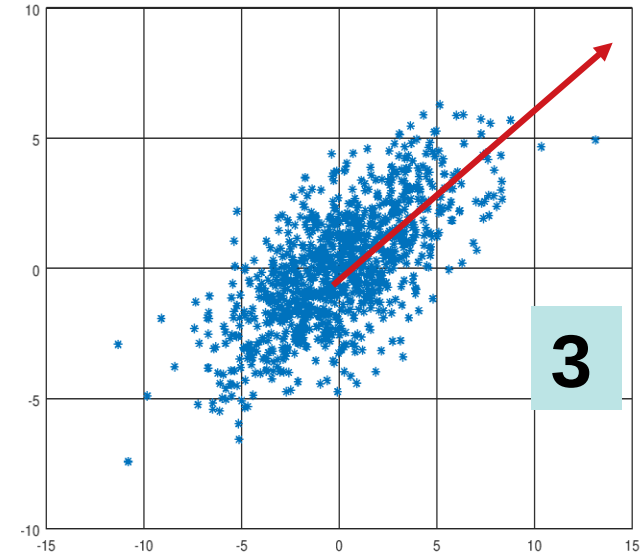
Properties of PCA



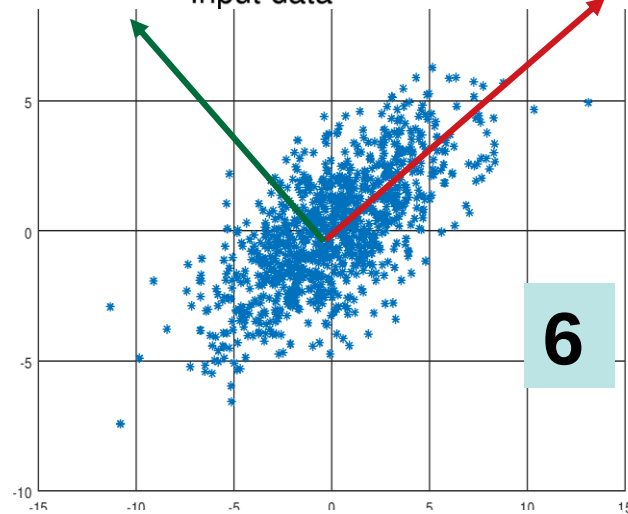
Input data



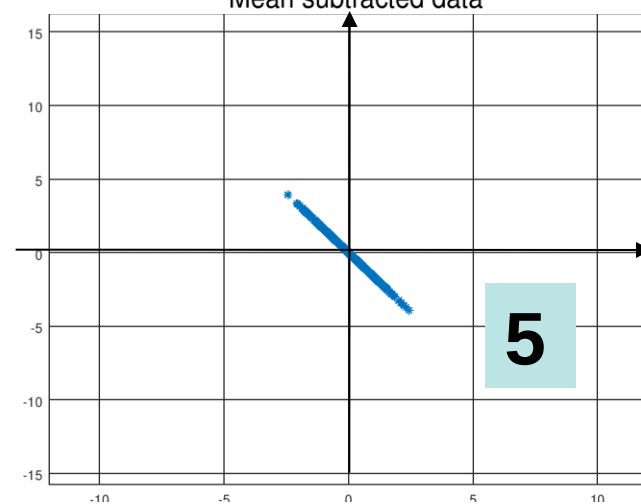
Mean subtracted data



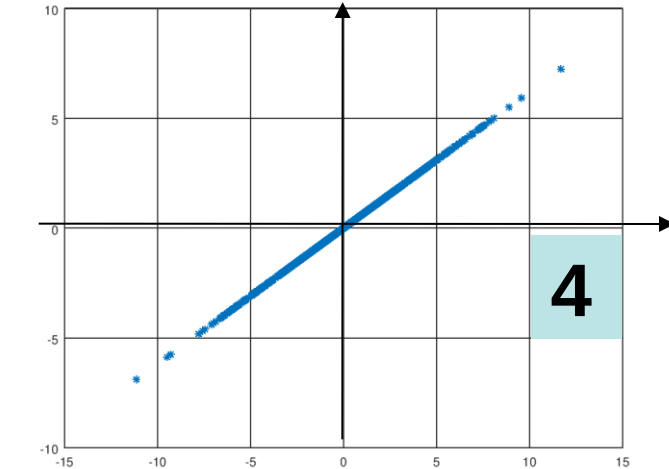
Axis of maximum variance



First and second axes of max. variance



Removal of first component (MeanSubData – Proj.)



Projection along Axis of maximum variance

Deriving solution to PCA

$$\begin{aligned} \mathbf{w}_1^* &= \underset{\|\mathbf{w}_1\|=1}{\operatorname{arg\,max}} \left[\sum_{i=1}^N X_{i_{new}}^2 \right] \\ &= \underset{\|\mathbf{w}_1\|=1}{\operatorname{arg\,max}} \left[\sum_{i=1}^N (X_i \mathbf{w}_1)^2 \right] \\ &= \underset{\|\mathbf{w}_1\|=1}{\operatorname{arg\,max}} \left[(X_1 \mathbf{w}_1)^2 + (X_2 \mathbf{w}_1)^2 + \dots + (X_n \mathbf{w}_1)^2 \right] \\ &= \underset{\|\mathbf{w}_1\|=1}{\operatorname{arg\,max}} \|\mathbf{X} \mathbf{w}_1\|^2 \\ &= \underset{\|\mathbf{w}_1\|=1}{\operatorname{arg\,max}} \left[(\mathbf{X} \mathbf{w}_1)^T \mathbf{X} \mathbf{w}_1 \right] \\ &= \underset{\|\mathbf{w}_1\|=1}{\operatorname{arg\,max}} \left[\mathbf{w}_1^T \mathbf{X}^T \mathbf{X} \mathbf{w}_1 \right] \end{aligned}$$

Note that this assumes that the data is mean zero

Deriving solution to PCA

$$\mathbf{w}_1^* = \underset{\|\mathbf{w}_1\|=1}{\mathit{arg\,max}} \left[(\mathbf{X} \mathbf{w}_1)^T \mathbf{X} \mathbf{w}_1 \right] = \underset{\|\mathbf{w}_1\|=1}{\mathit{arg\,max}} \left[\mathbf{w}_1^T \mathbf{X}^T \mathbf{X} \mathbf{w}_1 \right]$$

To solve the above maximisation we form the *Lagrangian*:

$$L(\mathbf{w}_1) = \mathbf{w}_1^T \mathbf{X}^T \mathbf{X} \mathbf{w}_1 - \lambda_1 (\mathbf{w}_1^T \mathbf{w}_1 - 1)$$

At the solution/saddle points the partial derivative will be 0. Hence:

$$\frac{\partial L(\mathbf{w}_1)}{\partial \mathbf{w}_1} = 2 \mathbf{w}_1^T \mathbf{X}^T \mathbf{X} - 2 \lambda_1 \mathbf{w}_1^T = \mathbf{0}$$

$$\mathbf{w}_1^T \mathbf{X}^T \mathbf{X} = \lambda_1 \mathbf{w}_1^T$$

$$(\mathbf{w}_1^T \mathbf{X}^T \mathbf{X})^T = (\lambda_1 \mathbf{w}_1^T)^T$$

$$\mathbf{X}^T \mathbf{X} \mathbf{w}_1 = \lambda_1 \mathbf{w}_1$$

Thus we conclude that $\mathbf{w}_1$ is an *eigen vector* for $\mathbf{X}^T \mathbf{X}$ and λ_1 is the corresponding *eigen value*.

Deriving solution to PCA

If $\mathbf{w}_2$ is another eigen vector with corresponding λ_2 eigen value then:

$$\mathbf{X}^T \mathbf{X} \mathbf{w}_1 = \lambda_1 \mathbf{w}_1$$

$$\mathbf{w}_2^T (\mathbf{X}^T \mathbf{X} \mathbf{w}_1) = \mathbf{w}_2^T (\lambda_1 \mathbf{w}_1)$$

$$(\mathbf{w}_2^T \mathbf{X}^T \mathbf{X}) \mathbf{w}_1 = \lambda_1 \mathbf{w}_2^T \mathbf{w}_1$$

$$(\mathbf{w}_2^T (\mathbf{X}^T \mathbf{X})^T) \mathbf{w}_1 = \lambda_1 \mathbf{w}_2^T \mathbf{w}_1 \quad \text{since the covariance matrix is symmetric}$$

$$(\mathbf{X}^T \mathbf{X} \mathbf{w}_2)^T \mathbf{w}_1 = \lambda_1 \mathbf{w}_2^T \mathbf{w}_1$$

$$(\lambda_2 \mathbf{w}_2)^T \mathbf{w}_1 = \lambda_1 \mathbf{w}_2^T \mathbf{w}_1 \quad \text{since } \mathbf{w}_2 \text{ is an eigen vector}$$

$$\lambda_2 \mathbf{w}_2^T \mathbf{w}_1 - \lambda_1 \mathbf{w}_2^T \mathbf{w}_1 = 0$$

$$(\lambda_2 - \lambda_1) \mathbf{w}_2^T \mathbf{w}_1 = 0$$

$$\text{Since } \lambda_2 \neq \lambda_1, \text{ hence } \mathbf{w}_2^T \mathbf{w}_1 = 0$$

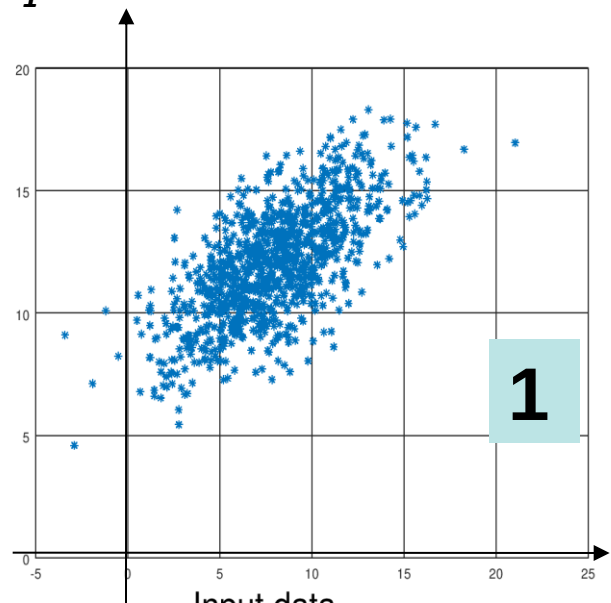
This demonstrates that all eigen vectors are (automatically) orthogonal.

Thus it is sufficient to compute the eigen vectors without needing to ensure orthogonality as an additional constraint in our algorithm.

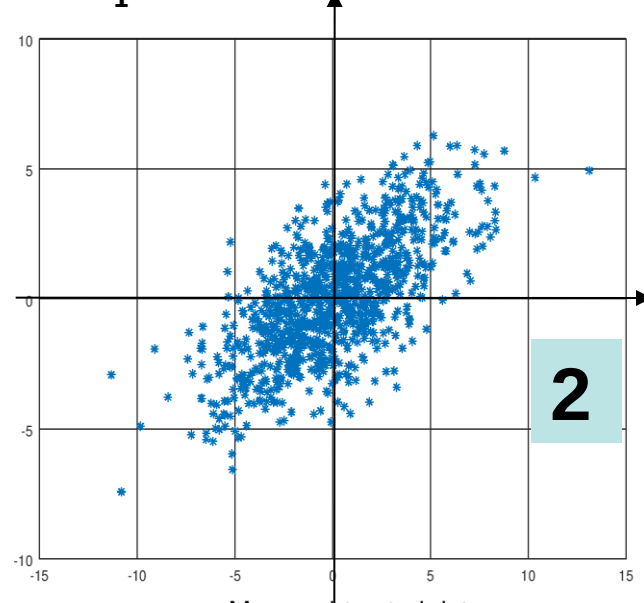
In other words, solution to PCA can be obtained by eigen decomposition of the covariance matrix.

Properties of PCA

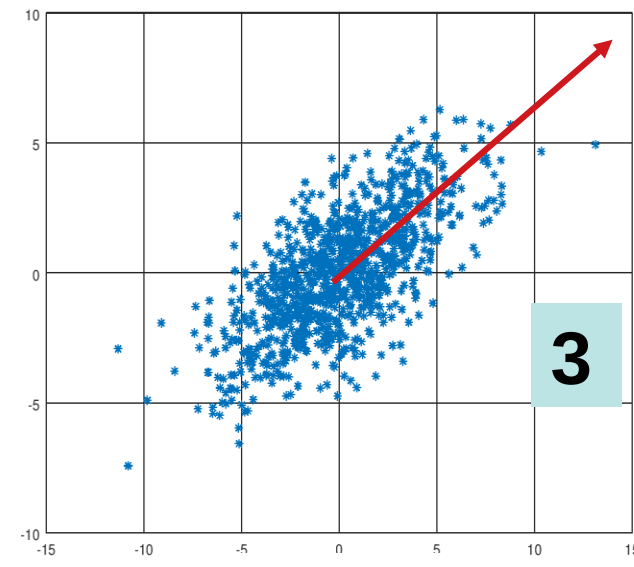
W_1 is the **largest eigen vector** for $X^T X$ and λ_1 is the corresponding **largest eigen value**.



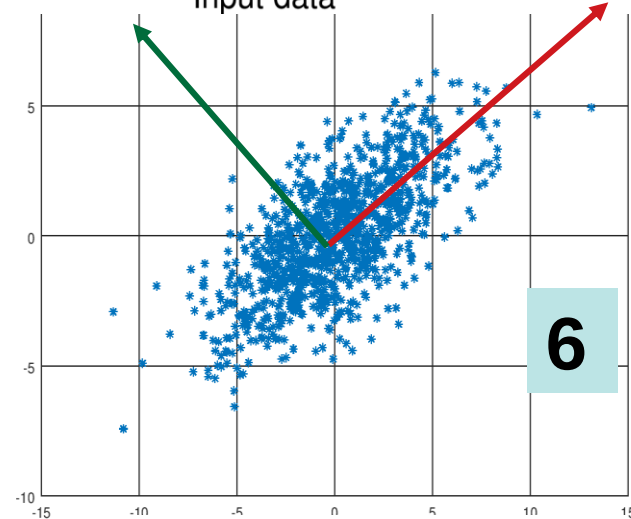
Input data



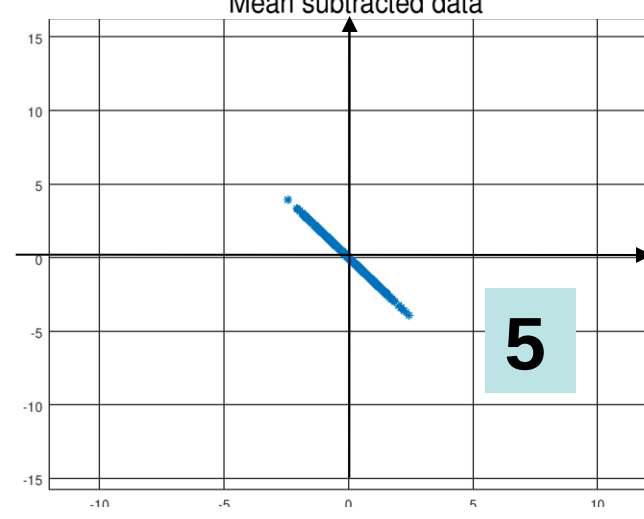
Mean subtracted data



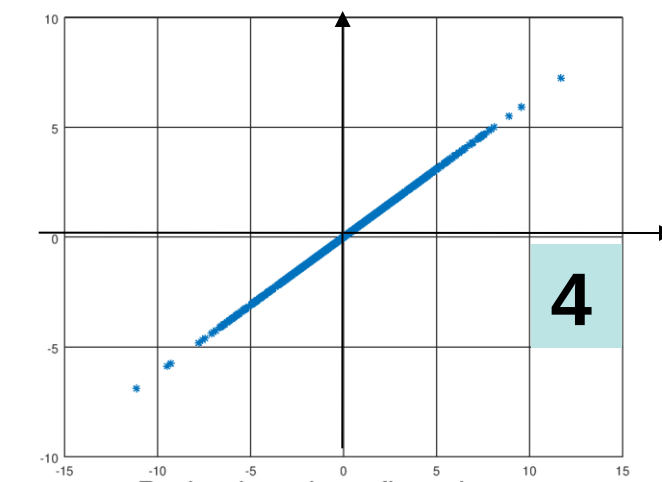
First eigen vector



First and second eigen vectors



Removal of first component (MeanSubData – Proj.)



Projection along first eigen vector

Recovering remaining components

- $\mathbf{w}_1$ is the **largest eigen vector** for $\mathbf{X}^T \mathbf{X}$ and λ_1 is the corresponding **largest eigen value**.
- We can find the second axis or the second eigen vector by removing the projection of the data points into the first eigen vector and then finding the next largest eigen vector.

- Projection of data point $\mathbf{x}$ along $\mathbf{w}_1$ is given by: $\mathbf{x} \mathbf{w}_1$ (scalar) ($\mathbf{x}$ is row v. and $\mathbf{w}_1$ is column v.)

- Projection of all data points along $\mathbf{w}_1$ is given by: $\mathbf{X} \mathbf{w}_1$

- Data point $\mathbf{x}$ along $\mathbf{w}_1$ is represented by: $(\mathbf{x} \mathbf{w}_1) \mathbf{w}_1^T = \mathbf{x} \mathbf{w}_1 \mathbf{w}_1^T$

- All data points along $\mathbf{w}_1$ is represented by: $\mathbf{X} \mathbf{w}_1 \mathbf{w}_1^T$

- Subtracting the first component from data gives: $\mathbf{Data}_{-1} = \mathbf{Data} - \mathbf{X} \mathbf{w}_1 \mathbf{w}_1^T$

- We can remove the first k components from the data by: $\mathbf{Data}_{-k} = \mathbf{Data} - \sum_{i=1}^k \mathbf{X} \mathbf{w}_i \mathbf{w}_i^T$

Properties of eigen vectors

For any symmetric $k \times k$ -dimensional matrix D (e.g. the covariance matrix $X^T X$)

- if w_1 is an eigen (column) vector and λ_1 is the corresponding eigen value then:

$$D w_1 = \lambda_1 w_1$$

- By the *spectral theorem*, all symmetric matrices have an eigen decomposition
- For an eigen decomposition of D , we have corresponding k eigen vectors such that:

$$D w_i = \lambda_i w_i \text{ for } 1 \leq i \leq k$$

- We can write the above compactly as:

- $D [w_1, \dots, w_k] = [\lambda_1 w_1, \dots, \lambda_k w_k]$

- $DW = W \Lambda$ where $\text{diag}(\Lambda) = [\lambda_1, \dots, \lambda_k]$:

$$\text{e.g. } W \Lambda = \begin{bmatrix} w_{11} & w_{21} & w_{31} & w_{41} \\ w_{12} & w_{22} & w_{32} & w_{42} \\ w_{13} & w_{23} & w_{33} & w_{43} \\ w_{14} & w_{24} & w_{34} & w_{44} \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_1 & 0 & 0 \\ 0 & 0 & \lambda_1 & 0 \\ 0 & 0 & 0 & \lambda_1 \end{bmatrix} = [\lambda_1 w_1 \quad \lambda_2 w_2 \quad \lambda_3 w_3 \quad \lambda_4 w_4]$$

- Thus: $D = W \Lambda W^T$

- And: $W^T D W = \Lambda$

A compact notation for
describing all eigen vectors

PCA Summary

Definition: The PCA problem can be defined formally as finding solution to the following problem:

$$\mathbf{W}^* = \underset{\mathbf{W}^T \mathbf{W} = \mathbf{I}}{\mathbf{arg\ max}} \mathbf{W}^T \mathbf{X}^T \mathbf{X} \mathbf{W}$$

where $\mathbf{I}$ denotes the identity matrix.

Solving PCA: The solution to PCA can be obtained by the eigen decomposition of the covariance matrix $\mathbf{X}^T \mathbf{X}$ given by:

$$\mathbf{X}^T \mathbf{X} = \mathbf{W} \mathbf{\Lambda} \mathbf{W}^T$$

where the column vectors of $\mathbf{W}$ are the orthonormal eigen vectors and $\mathbf{diag}(\mathbf{\Lambda})$ contains the corresponding eigen values.

Iterative algorithm for computing all eigen vectors

This suggests an iterative algorithm for computing eigen vectors

Algorithm sketch:

- Let $\mathbf{X}$ = *mean subtracted data*
- Let $\mathbf{Z} = \mathbf{X}$ (*copy of $\mathbf{X}$*)
- For $j = 0$ to k where k is desired number of eigen vectors ($k < d$):
 - Compute covariance matrix $\mathbf{Z}^T \mathbf{Z}$
 - Compute the largest eigen vector $\mathbf{w}_j$ of $\mathbf{Z}^T \mathbf{Z}$
 - Remove the eigen vector projections from data

$$\mathbf{Z} = \mathbf{X} - \sum_{i=1}^j \mathbf{X} \mathbf{W}_i \mathbf{W}_i^T$$

- Return the k eigen vectors

Points to note:

- This is not a practical algorithm since both steps:
 - recomputing covariance matrix and
 - largest eigen vector is expensive

are expensive

- Nevertheless, this is very useful to understand what we are aiming to do

Computing largest eigen vector

- We are interested in finding the largest eigen vector $\mathbf{w}$ given by: $\mathbf{w} \lambda = \mathbf{X}^T \mathbf{X} \mathbf{w}$

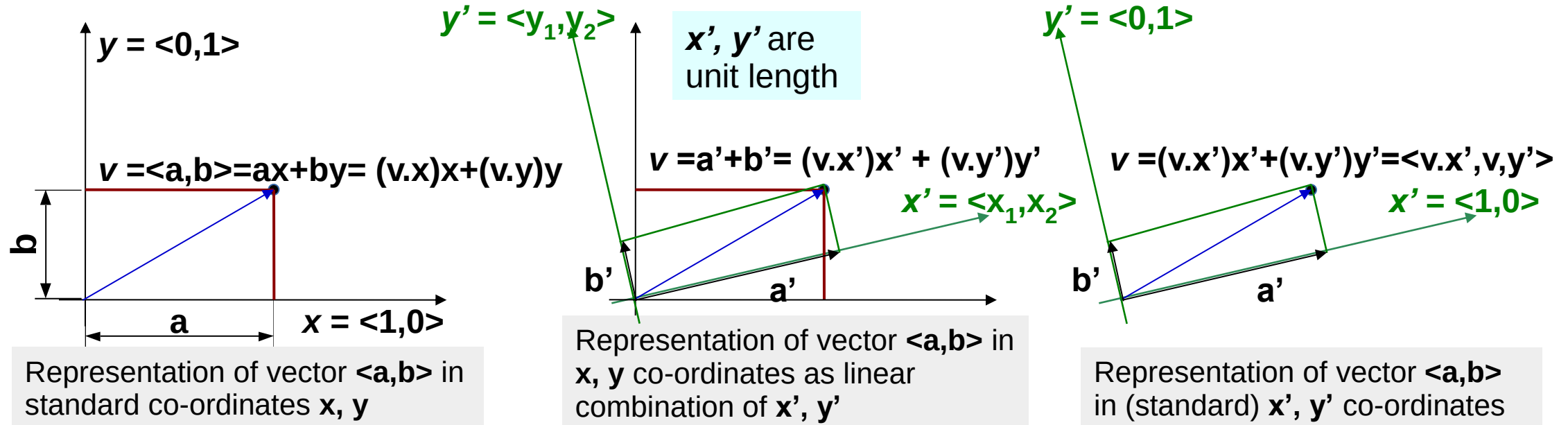
such that $||\mathbf{w}|| = 1$

- If the eigen values are sufficiently separated, then a simple algorithm can find the largest eigen value

Algorithm (Power iteration):

- Initialise $\mathbf{w}$ as a random vector
- $\mathbf{w} = \mathbf{w} / ||\mathbf{w}||$
- Compute $\mathbf{X}^T \mathbf{X}$
- While not termination criteria:
 - $\mathbf{w} = \mathbf{X}^T \mathbf{X} \mathbf{w}$
 - $\mathbf{w} = \mathbf{w} / ||\mathbf{w}||$
- Method will only work if the eigen vector is sufficiently dominating
i.e. the eigen value is sufficiently larger than the next largest eigen value.

Representation using new bases



- So input data point $v = \langle a, b \rangle$ becomes $\langle v.x', v.y' \rangle$ in the new co-ordinate system
- $v = \langle a, b \rangle = (v.x')x' + (v.y')y' = \alpha_1 x' + \alpha_2 y'$ where $\alpha_1 = v.x'$ and $\alpha_2 = v.y'$
- This means, if matrix W contains the eigen vectors w_i (as columns) and x is a data point then:

$$x = \sum_i \alpha_i w_i^T = \sum_i x w_i w_i^T$$

- In the new eigen bases this becomes: $x_{new} = \langle \alpha_1, \dots, \alpha_D \rangle = \langle x w_1, \dots, x w_D \rangle = x W$
- Continuing from previously we note that:

$$x = \sum_i \alpha_i w_i^T = \sum_i x w_i w_i^T = x \sum_i w_i w_i^T = x W W^T = x I = x$$

Eigen decomposition algorithms

- Here we will simply note that efficient methods for are available.
 - Specialised PCA algorithms are further optimised so that the covariance matrix is not explicitly precomputed
 - Similarly, efficient algorithms for eigen decomposition are available. For doing PCA this will require precomputing the covariance matrix.
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- **Practicals** – we will use black box solutions either for PCA or eigen decomposition of the covariance matrix

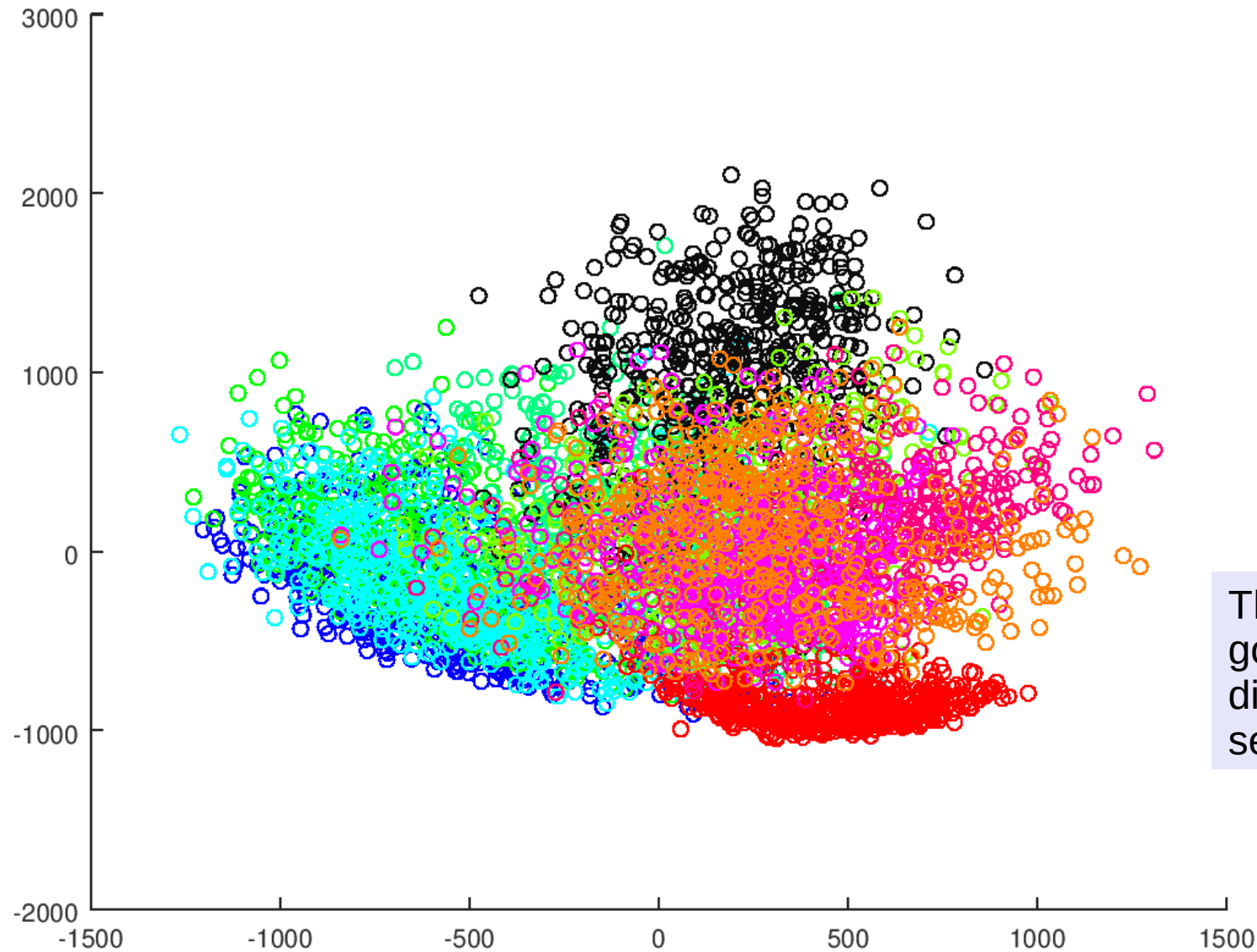
Dimensionality reduction

Purpose: find a low dimensional representation that describes the data succinctly
i.e. using a few number of features

Method:

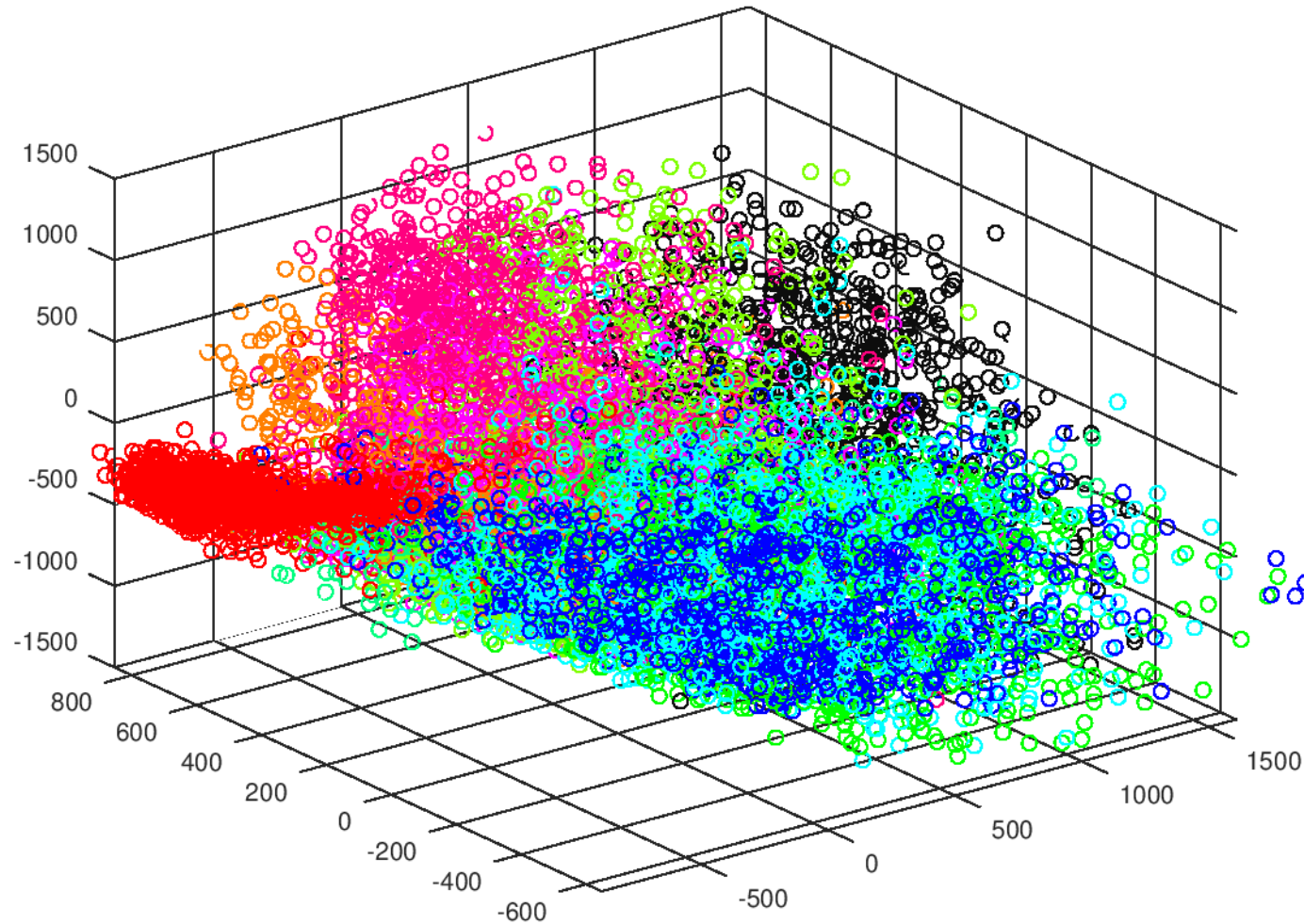
- Either use a specialised PCA algorithm
- Or perform eigen decomposition of the covariance matrix
 - Sort the eigen values in decreasing order
 - Choose the top **k** eigen vectors as the new bases
- To choose **k**, one can use the following method:
 - Compute running total of all eigen values so far i.e.:
 - $T = 0$
 - $T = T + \text{eigen value}$
 - If $(\text{new eigen value} / T) < \text{Threshold (eg. 10\%)}$ then stop

Example (MNIST digits – 2 bases)



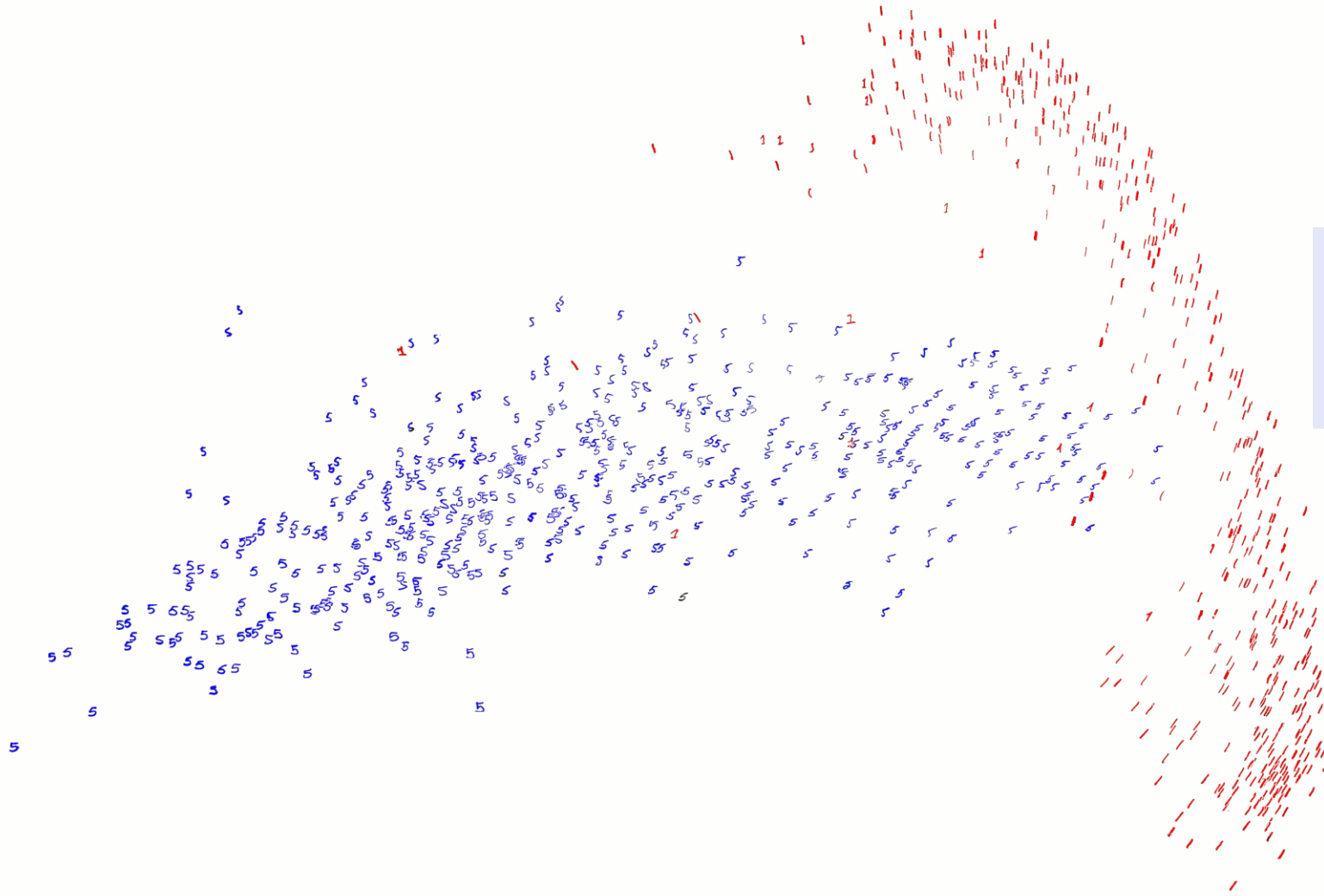
The first two eigen bases show good separation for some of the digits but not all digits are clearly separated in two dimensions

Example (MNIST digits – 2 bases)



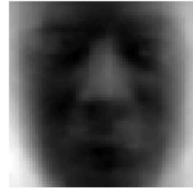
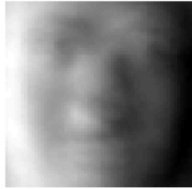
Does not show good linear separation of all digits. But polynomial features shows promise.

Example (MNIST digits – 2 bases)



Good separation for digits 1 and 5 only

Example (Top 10 Eigen Faces)



Each successive eigen face represents the most significant discriminating pixels



Summary

- PCA is a linear dimensionality reduction method
- It requires data to be mean centred
- PCA is equivalent to generating an orthogonal (unitary) transformation of the bases i.e. rotation, reflection + combinations
- PCA can be done by computing the eigen decomposition of the covariance matrix
- Simple (but inefficient) iterative methods exists for computing eigen vectors
- Can scale to large datasets with efficient methods